Introduction to Machine Learning

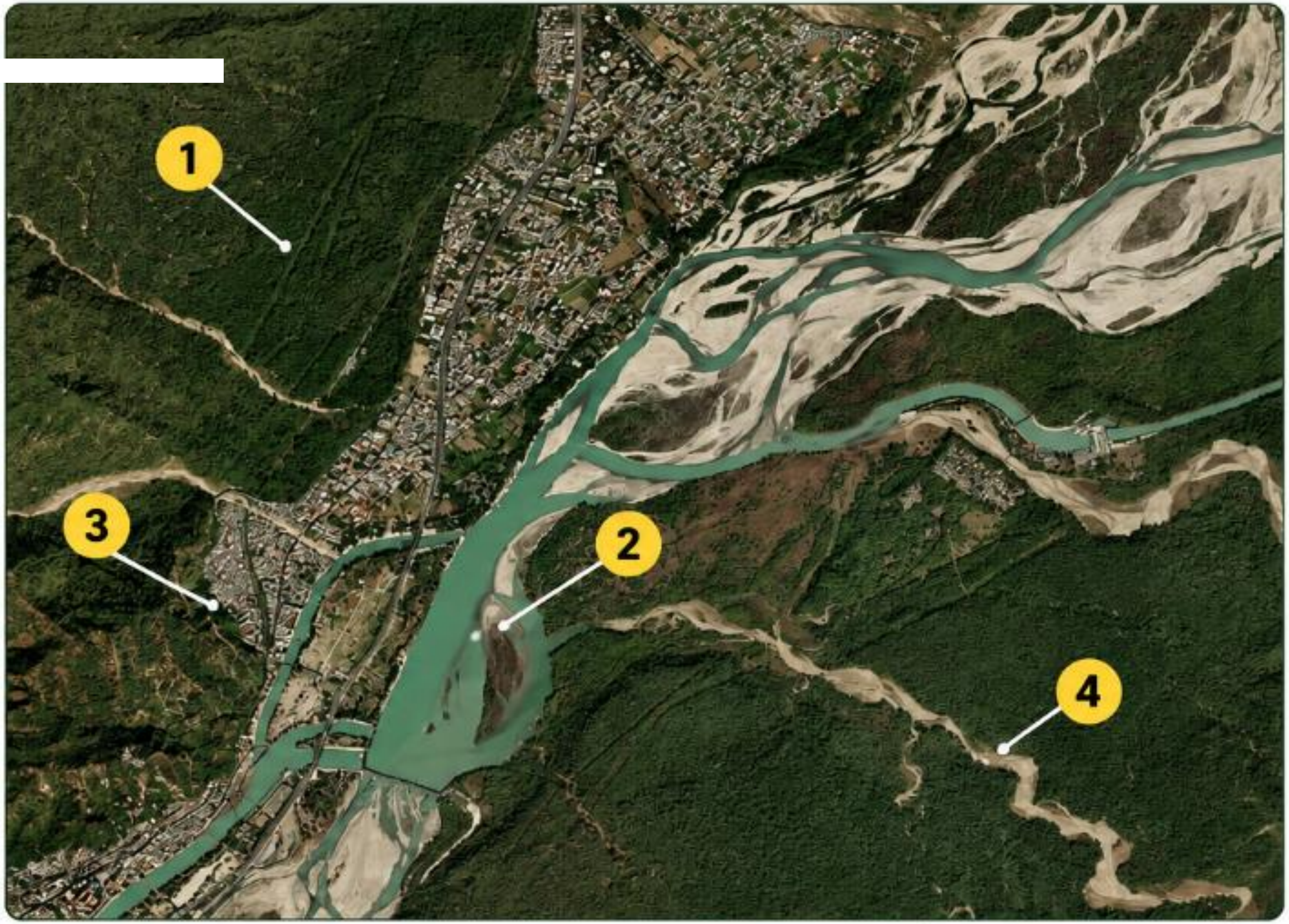
From Basic Concepts to Decision Trees &Random Forests

Can a Computer Learn This?

Look at this satellite image.
What do you see?

Think for a moment!
What do you think the numbered areas represent?

1. What is area 1?
2. What is area 2?
3. What is area 3?
4. What is area 4?

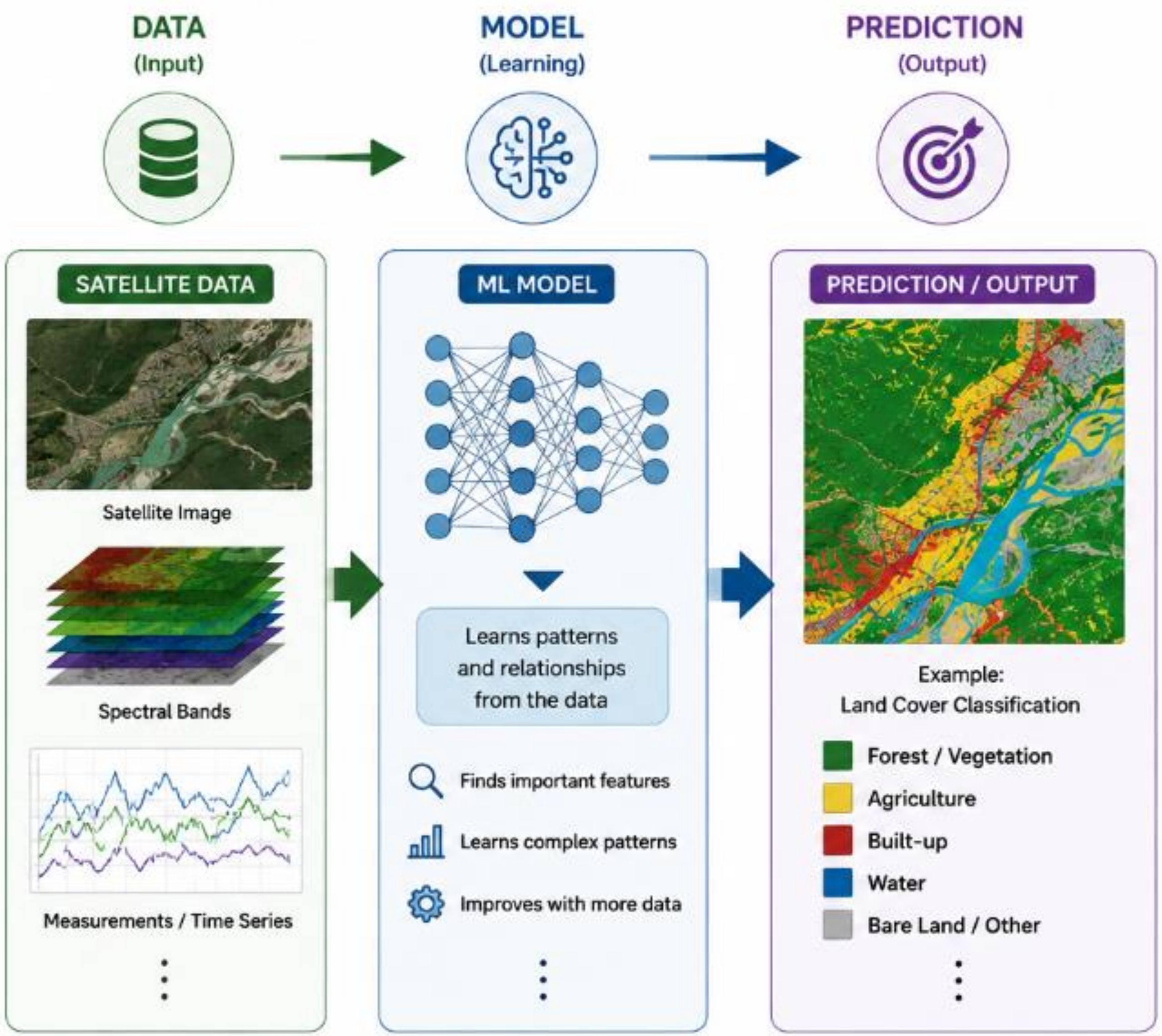


Can we teach a computer to identify these features automatically?

This is where Machine Learning comes in.

What is Machine Learning?

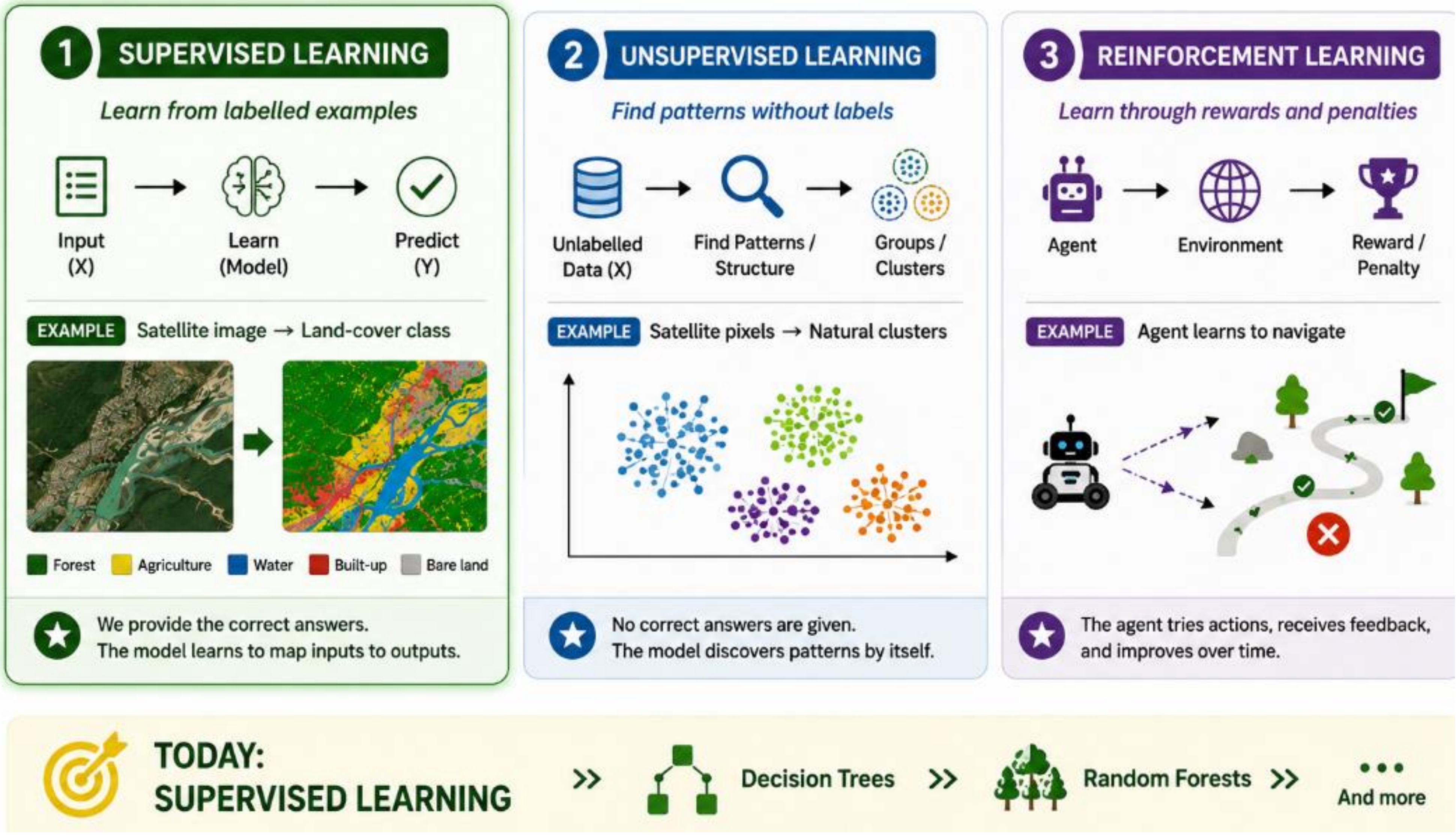
- **Main Point:** Teaching computers to learn patterns from data and use those patterns to make predictions or decisions.
- **Key Concepts:**
  - **Data:** The fuel for learning (numbers, text, images).
  - **Model:** The mathematical engine that finds patterns.
  - **Prediction:** Using the model to make decisions on new, unseen data.



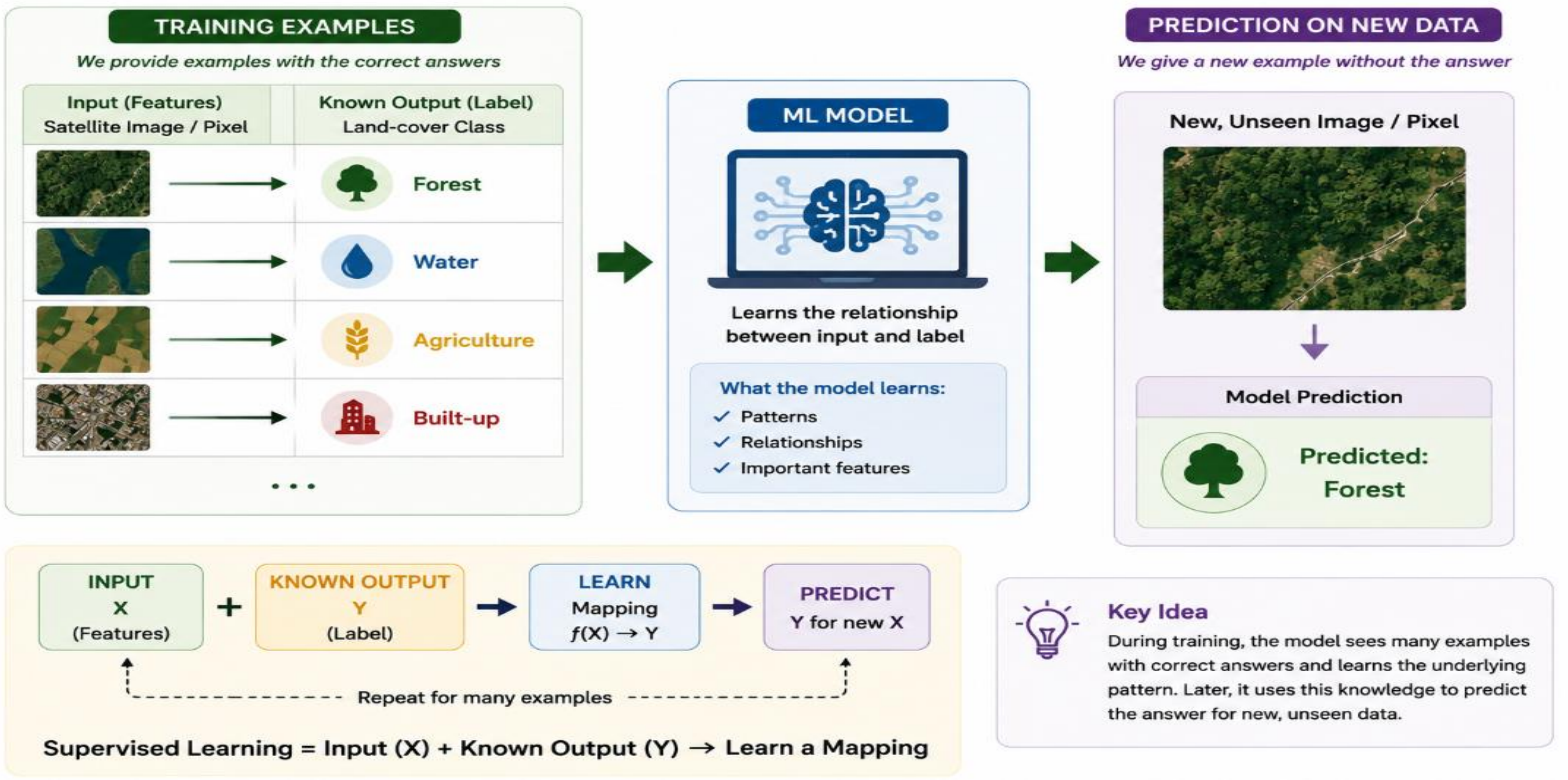
Machine Learning =
Learning patterns from data → Applying them to new data

Once trained, the model can make predictions on new, unseen data quickly and consistently.

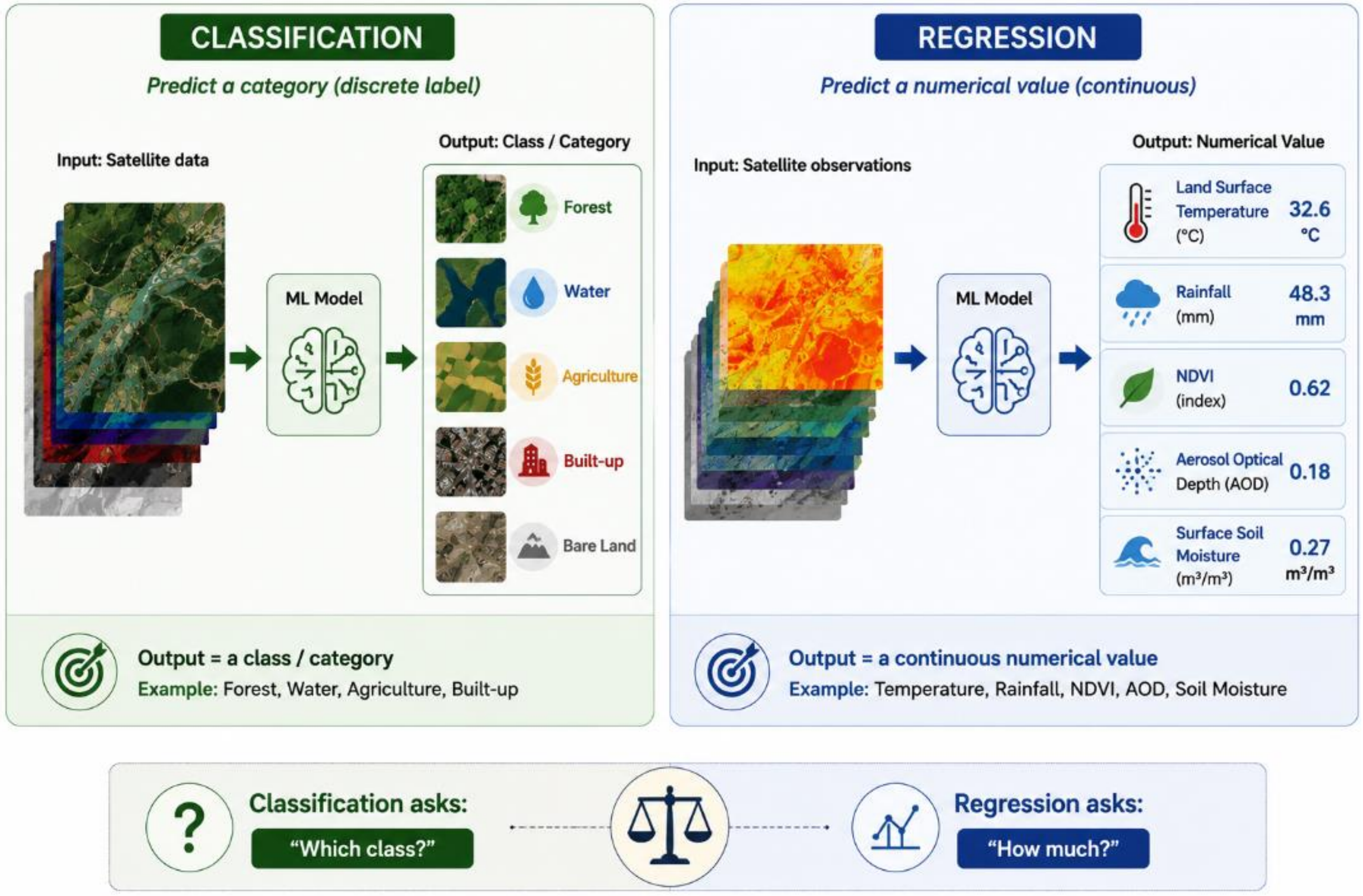
Types of Machine Learning



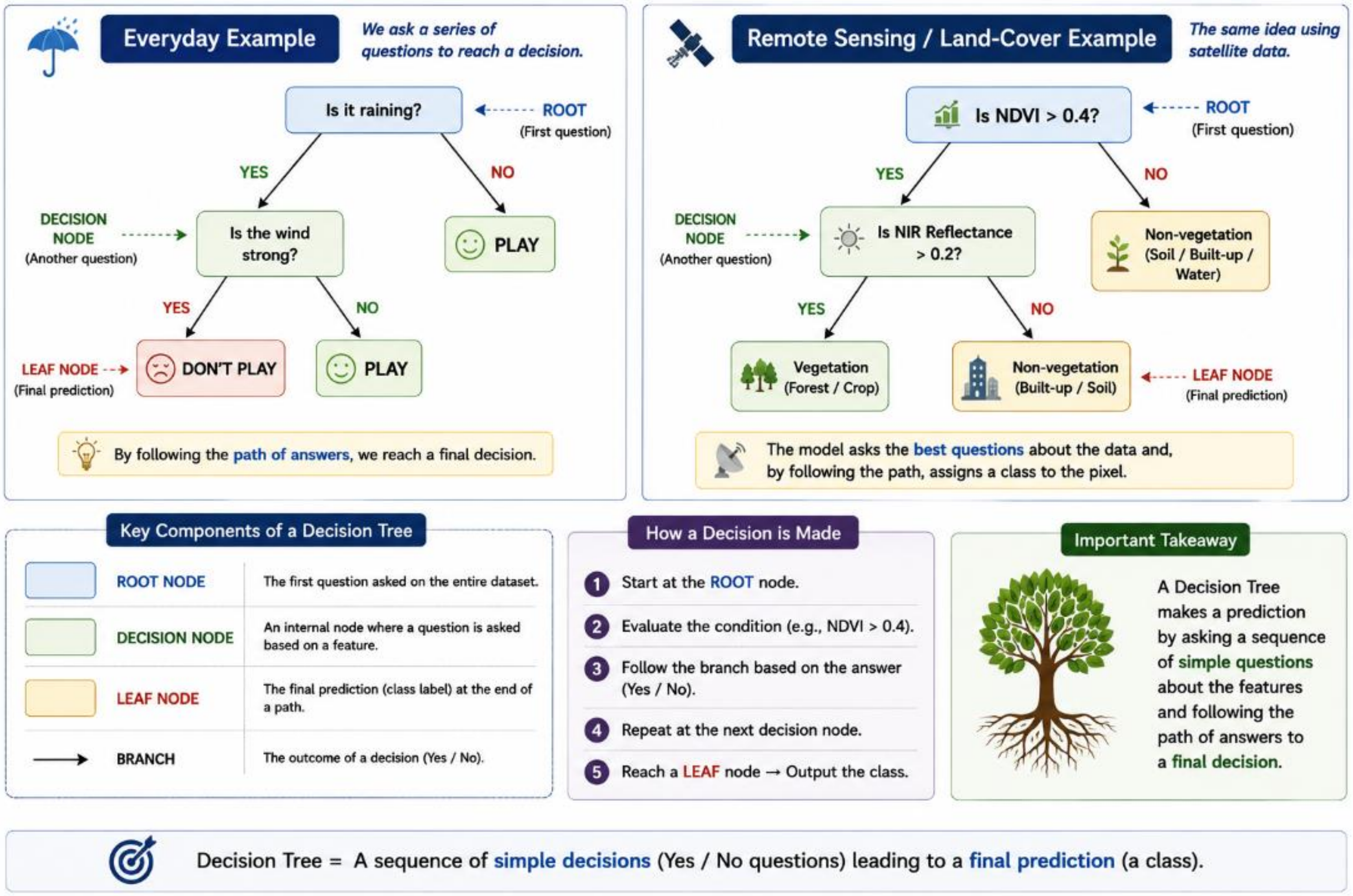
Supervised Learning : Learning from Examples



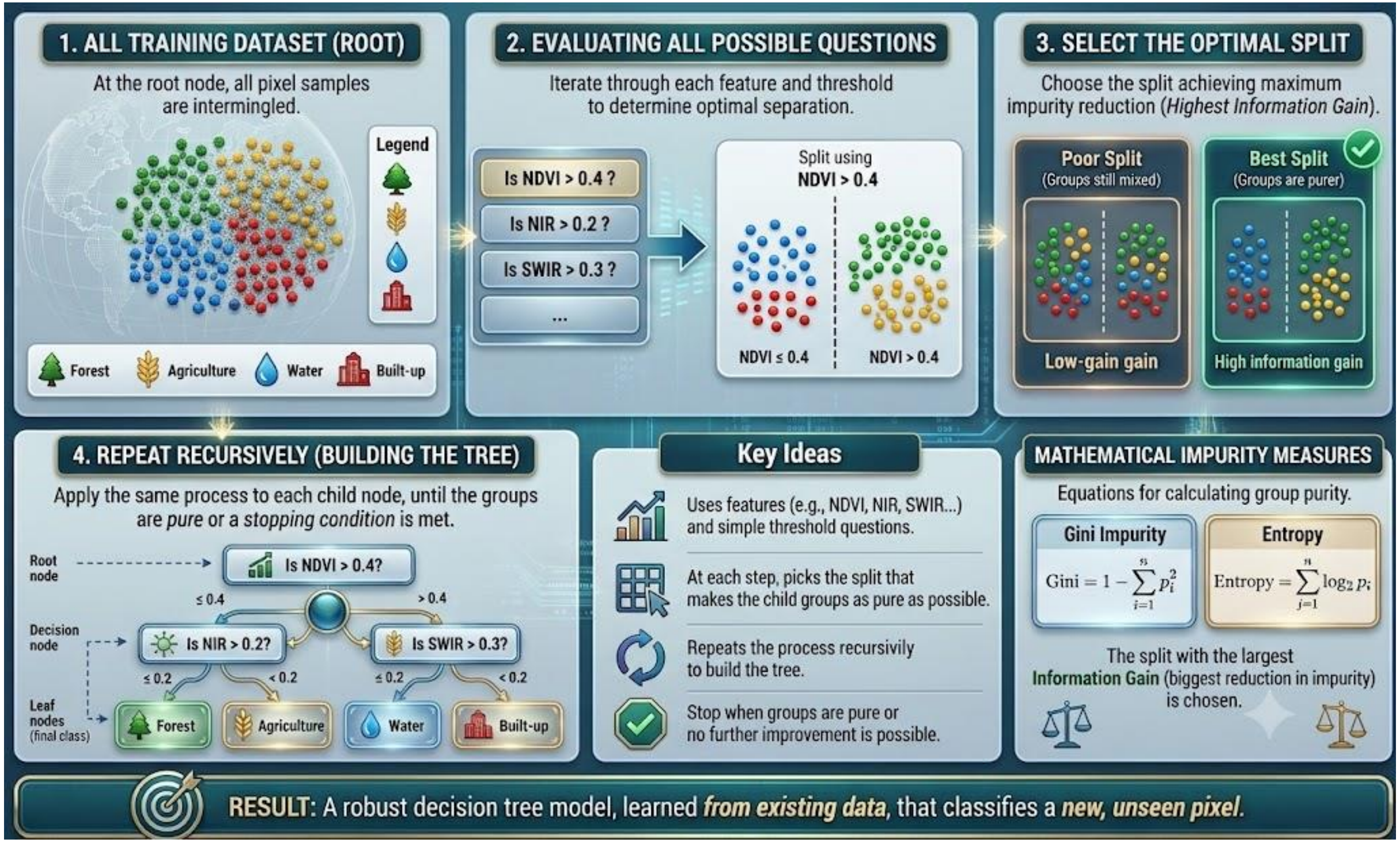
What Are We Trying to Predict?



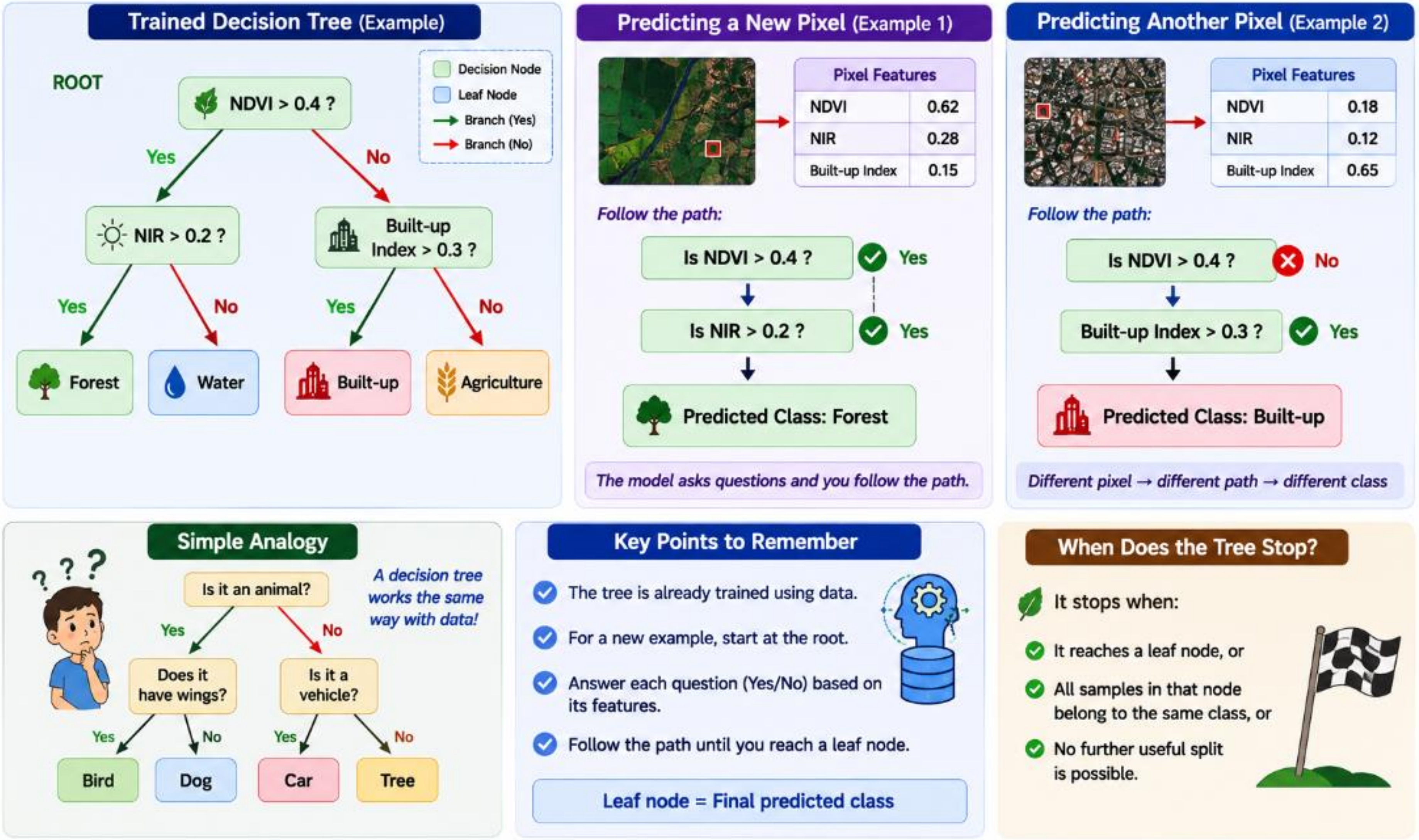
Decision Tree: A Sequence of Simple Decisions



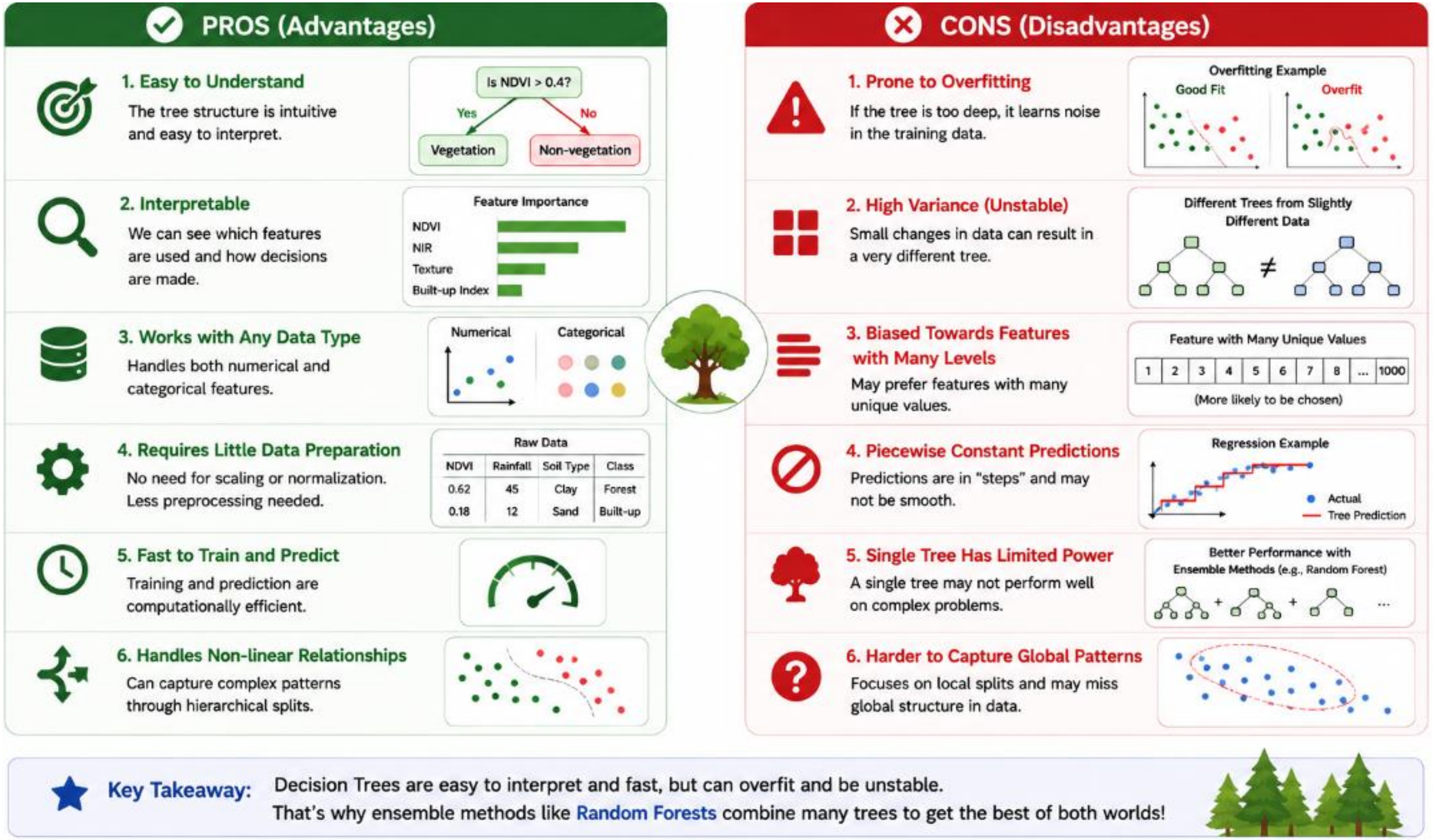
How Decision Tree Learn



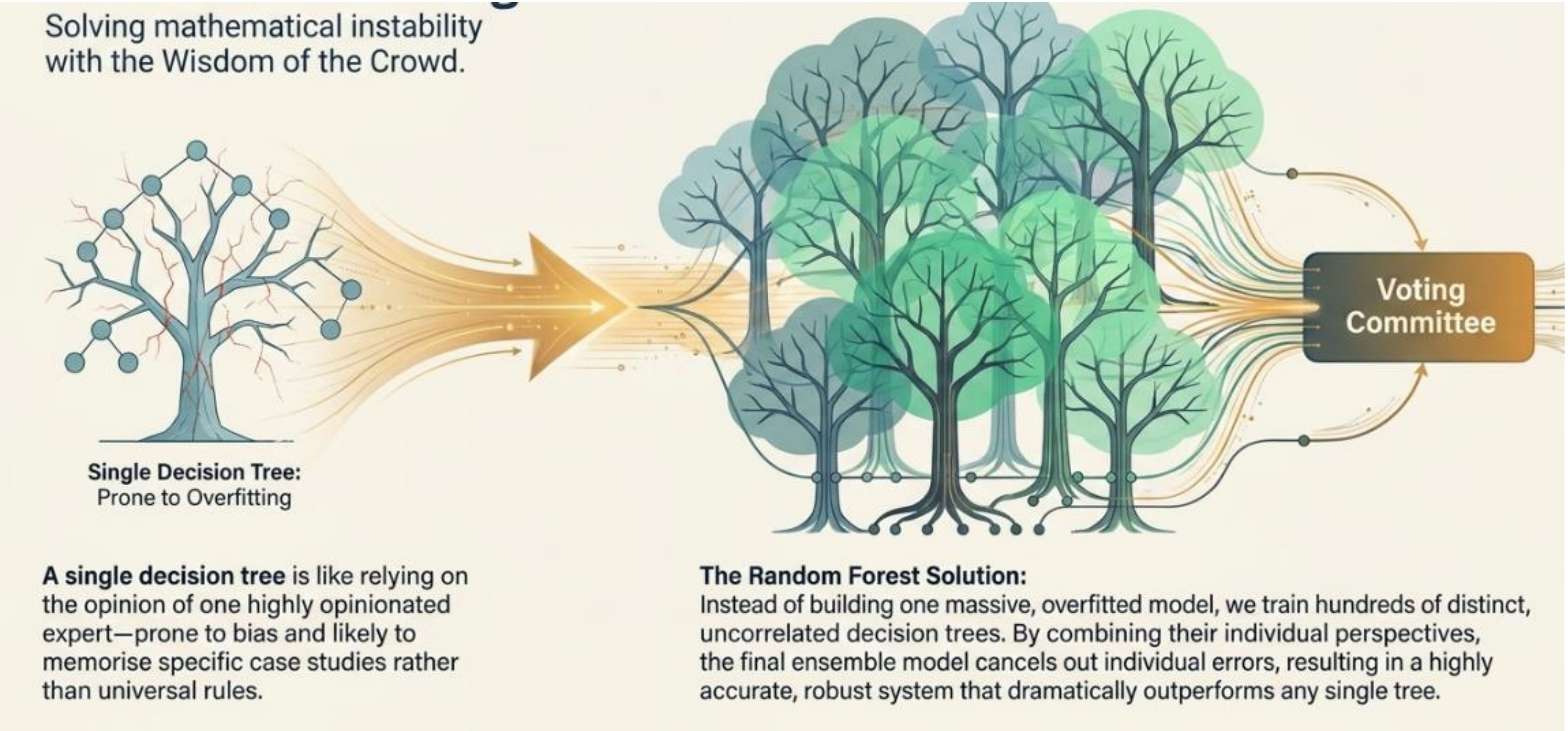
Using a Decision Tree



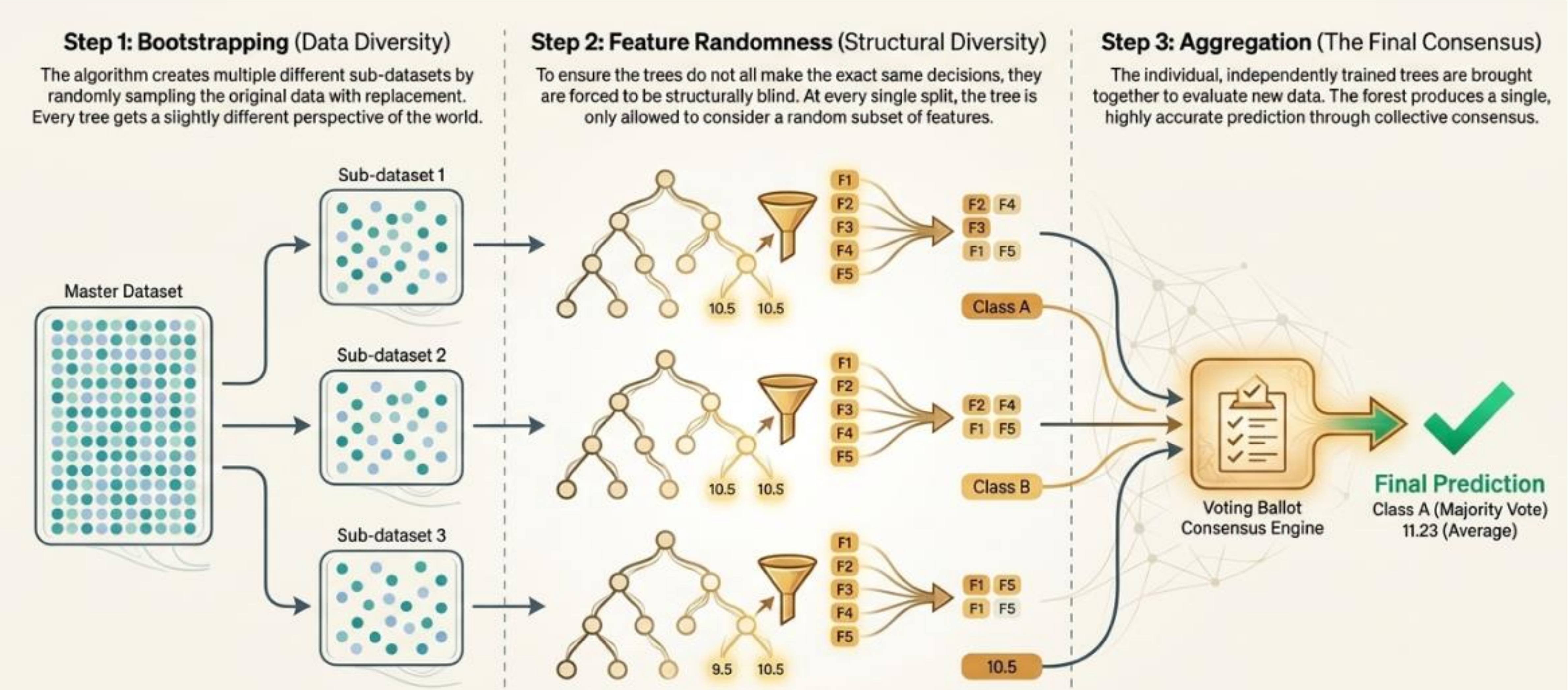
Pros & Cons of Decision Trees



Paradigm Shift: Ensemble Learning



Random Forest



Random Forest vs Decision Tree

Single Decision Tree	Random Forest Ensemble
✗ **Architecture:** One deeply grown model.	✓ **Architecture:** Hundreds of uncorrelated models acting together.
✓ **Interpretability:** Extremely high. Logic can be mapped visually.	✗ **Interpretability:** Low (a "Black Box" model). Difficult to trace exactly why a specific decision was made.
✗ **Performance:** Brittle. Highly prone to overfitting and memorising noise.	✓ **Performance:** Highly robust. Immune to minor variations in training data.
✓ **Speed:** Near-instantaneous training and prediction.	✗ **Speed:** Computationally heavy; requires more time and resources.
✓ **Best For:** Simple datasets datasets where explaining the decision is more important than absolute accuracy.	✓ **Best For:** Complex, high-stakes real-world environments where precision and reliability are non-negotiable.

Applications in Remote Sensing

Thank You